DESIGN COMPUTED · FREE GUIDE

Tune it, don't replace it: how Big Ben is kept on time with pennies

The free guide to the reel on the Great Clock of Westminster: how a pendulum keeps time, why a penny on it changes the rate, and how to build and tune your own pendulum with coins.



A clock run by a pendulum

The Great Clock in the Elizabeth Tower, designed by Edmund Beckett Denison and built by the firm of Dent, started in 1859. It is still mechanical: a pendulum almost 4 m long beats once every 2 seconds, and the gears count those beats to move the hands.

A pendulum's beat depends almost only on its length: the longer it is, the slower it swings. That is why it can keep good time, and also why a tiny change in length changes the time.

Why a penny works

On the pendulum sits a small tray. Putting an old penny on it moves the pendulum's centre of mass up very slightly. The pendulum then behaves as if it were a little shorter, so it swings a little faster.

One penny makes the clock gain about 0.4 seconds a day (two fifths of a second). Taking one off slows it by the same amount. No need to stop the clock or touch the hands.

Built examples of fine adjustment

Great Clock of Westminster, London	Pre-decimal pennies on the pendulum adjust the rate by about 0.4 s a day each.
Pendulum clocks in general	A rating nut under the bob raises or lowers it to set the rate; some precision clocks also use small added weights.
Tuned mass dampers (Taipei 101)	The same physics in a building: the length of the pendulum is chosen so it swings at the tower's own rhythm.

Try it yourself: tune a pendulum with coins

1. Tie a heavy nut or a bag of coins to a 1 m string and hang it from a door frame.
2. Pull it 10 cm aside, let go, and time 20 swings with your phone. Divide by 20 to get one swing.
3. Tape a small paper cup to the string about a third of the way down and repeat: note the time.
4. Put two coins in the cup and time 20 swings again. The pendulum is now slightly faster.

5 Move the cup higher or lower with the same coins and see how the position changes the effect. You are doing exactly what Big Ben's keepers do.

The rule to remember

Before replacing something that works, look for the smallest adjustment you can measure and undo.

Default answer	It drifts, so replace it with something new.
Form-finder answer	Find the one parameter that controls it and tune it in small steps.
Check	Ask: what is the penny here, the smallest change with a measurable effect?

Go further

UK Parliament: the Great Clock
https://www.parliament.uk/bigben

Big Ben (Wikipedia)
https://en.wikipedia.org/wiki/Big_Ben

Britannica: Big Ben
https://www.britannica.com/topic/Big-Ben-clock-London

Credits
Cover photo: Ryan Hodnett, CC BY-SA 4.0, Wikimedia Commons. Mechanism plan: E. B. Denison, c.1854, public domain.
Facts: UK Parliament, Britannica, BBC, Wikipedia.

Made by Design Computed · Follow for more · @designcomputed